

ÖNERİLEN NEXT.JS WEB PLATFORM MİMARİSİ

Multi-tenant · Cloud-native · Semantic Search · Provider-abstracted AI

CLIENT KATMANI — NEXT.JS 16 + REACT + TYPESCRIPT

/dashboard Legal Intelligence Dashboard Son araştırmalar, dilekçeler, hızlı işlemler	/search Semantic + Keyword Hybrid Search Benzerlik skoru · metadata filtre	/petitions Dilekçe oluştur + düzenle + versiyon PDF/DOCX export · diff	/research/[id] Research Trace UI Şeffaf AI adımları	/chat AI Chat (kaynak gösteren) SSE streaming · tool dispatch
/documents Kullanıcı dosya yükleme PDF/DOCX parse · OCR · embed				

↓ HTTPS / SSE / WebSocket

API GATEWAY — NEXT.JS API ROUTES (TYPESCRIPT)

/api/auth/* NextAuth.js + JWT + RBAC admin/lawyer/user/org_admin	/api/legal/search/* 5 search endpoint (bedesten/deep/mevzuat/aym/rg) Hybrid: BM25 + pgvector + rerank	/api/petitions/* generate · review · versions · compare · export WeasyPrint + python-docx	/api/research/* sessions · traces · steps Gerçek araştırma verisi	/api/documents/upload PDF/DOCX → parse → embed → index Async job queue
/api/admin/* User mgmt · audit logs · usage Rate limit config				

↓ Internal HTTP (synchronous) + Job Queue (async)

PYTHON LEGAL ENGINE SERVICE — INTERNAL FASTAPI CONTAINER

engine-wrapper Mevcut engine.py modüllerleştirilmiş hali research.py · petition.py · llm.py	mcp-adapter 21 MCP aracının FastAPI router'a çevrilmesi 18/21 araç doğrudan çevrilebilir	quality-engine quality.py + motor/pipeline.py (sync → threadpool) 17 boyut puanlama · halüsinasyon kontrol	petition-renderer WeasyPrint + python-docx + Jinja2 PDF/DOCX/HTML çıktı	source-connectors Async httpx + Redis rate-limiter 6 hukuk portalı · proxy desteği
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AI PROVIDER ABSTRACTION (YENİ)

OpenAI GPT-4o · GPT-4-Turbo · o1 AI_PROVIDER=openai	Google Gemini Gemini 2.0 Flash · Pro AI_PROVIDER=gemini	Anthropic Claude 3.5 Sonnet · Haiku AI_PROVIDER=anthropic	OpenRouter 100+ model tek API AI_PROVIDER=openrouter	Ollama / llama.cpp Qwen 2.5/3 (legacy uyumluluk) AI_PROVIDER=ollama
Router Provider seçici + fallback + retry env-driven · cost-aware				

HYBRID SEARCH ENGINE (YENİ)

Semantic Search multilingual-e5-large (1024d) veya OpenAI embed (1536d) pgvector + HNSW index	Keyword Search PostgreSQL FTS (tsvector) + BM25 rank Türkçe stopwords · stemming	Metadata Filter court · chamber · date · type · source Pre-filter before rerank	Reranker Cross-encoder (bge-reranker-tr) Top-50 → Top-10	Query Expansion AI tabanlı sinonim/kavram genişletme "güvenlik soruşturması" → "arşiv araştırması, atanma..."
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VERİTABANI — POSTGRESQL 16 + PGVECTOR

users · organizations Multi-tenant auth + RBAC subscriptions · api_keys	cases · case_parties Dava yönetimi + taraflar documents · citations	legal_decisions 9M+ içtihat (partitioned by year) court · chamber · decision_no	document_chunks Belgelerin parçalanmış hali chunk_text · metadata JSONB	document_embeddings pgvector 1536d · HNSW index cosine similarity
statutes · statute_articles Versiyonlu mevzuat (v1/v2/current) effective_from · effective_to		petitions · petition_versions Dilekçe + sürüm geçmişi diff · compare · rollback		research_sessions · traces AI araştırma şeffaflık kayıtları audit_logs · ai_runs

ASYNC JOB QUEUE — REDIS + BULLMQ

Import Pipeline 9M+ belge → parse → chunk → embed → index Queue → Worker → Vector DB	Source Sync Scheduled: Yargıtay/Danıştay/AYM/Mevzuat Changed? → Version → Re-embed	Document Upload User PDF/DOCX → OCR → embed User private vector index	AI Research Long-running research sessions SSE progress streaming
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STORAGE & INFRASTRUCTURE

S3-Compatible PDF/DOCX dosya depolama MinIO veya AWS S3	Redis Cache + Rate limit + Session Source portal 429 cooldown	Vercel / Docker Next.js frontend + API routes Edge functions mümkün	Python Container FastAPI internal service Docker · k8s · horizontal scale	Observability Sentry + OpenTelemetry + Grafana Audit logging · metrics
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İletişim Akışı: Browser → Next.js API routes → (sync) Python service OR (async) Job Queue → PostgreSQL + pgvector

Multi-tenant: Her org kendi cases/petitions/documents · Legal sources global · Embeddings user-private + global

Yeni Kabiliyetler: Semantic search · Çok-kullanıcılı · Provider abstraction · Versiyonlu mevzuat · Research trace · Document upload

Korunan Değerler: Mevcut 21 MCP aracı · 17-boyut kalite kontrol · 10 dava türü tespiti · Halüsinasyon kontrolü · 12 dilekçe şablonu